WEEK 3 PROGRESS REVIEW

Document Intelligence Benchmarking

OCR, Table Extraction & Layout Detection — Results Review

LAST WEEK

- benchmarking on market standard datasets
- GT and it’s bboxes generated manually by custom made website
- Benchmarking on layout models

THIS WEEK

- Corrected GT tightness for fair geometric scoring(via synthetic dataset generation)
- Benchmarking on this dataset for layout, text and table extraction

METHODOLOGY

The Synthetic Dataset Generation Pipeline

1
HTML / Jinja Templates
LLM-assisted catalog layout design

→

2
Python Faker Population
Realistic fields — forceps, retractors, scalpels, scissors

→

3
Playwright DOM Query
getBoundingClientRect() during render

→

4
PDF Compile
Zero-error, pixel-perfect ground truth

Zero-Error Ground Truth: bounding boxes extracted directly from the browser DOM at render time — no manual labeling, no human annotation error.(Generated by Claude Sonnet-5 Model)

Addressing the Drawbacks of Open-Source Datasets

Limitation of Open Datasets	Our Generator
Generic academic text / placeholder blocks	Realistic catalog copy, prices & specs with themed styling
No product image generation	Realistic generated product photo assets
Repetitive, duplicated page layouts	Randomized columns, element heights & row counts

Layout Detection: Models & Methodology

DocLayoutYOLO — Lightweight, CPU-capable region detector

Nemotron-Parse-v1.1 — ViT-H encoder + mBART decoder multimodal parser

Layer 1 · Geometric

- Precision, Recall, F1, Mean IoU, Class Accuracy
- Matching: Hungarian algorithm at IoU ≥ 0.5
- Higher IoU = tighter box alignment to GT

Layer 2 · Spatial Quality (COTe)

- Coverage, Overlap, Trespass, Excess
- Pixel-mask based evaluation of region fill
- Low Overlap / Trespass = cleaner region separation

Layer 3 · Layout Edit Distance (LED)

- Error categories: Missing, Hallucination, Size-Error, Split, Misclassification
- Counts edits needed to reconcile prediction with GT
- Lower counts = fewer structural errors

Layer 4 · Reading Order

- ROKT (Kendall's τ) and ROA on document flow
- τ = 1.0 indicates perfect predicted reading sequence

LAYOUT BENCHMARK · GT-TIGHT

Layout Detection: Results & Interpretation

Layer	Metric	DocLayoutYOLO	Nemotron-Parse
L1 Geometric	Precision	0.2381	0.2712
L1 Geometric	Recall	0.4517	0.4101
L1 Geometric	F1-Score	0.3072	0.3251
L1 Geometric	Mean IoU	0.8060	0.7464
L1 Geometric	Class Accuracy	53.04%	86.61%
L2 COTe	Coverage	0.6059	0.6575
L2 COTe	Overlap	0.0000	0.0868
L2 COTe	Trespass	0.0156	0.0233
L2 COTe	Excess	0.6157	0.6732
L3 LED	Missing	3.00	1.00
L3 LED	Hallucination	18.00	6.00
L3 LED	Size-Error	7.00	10.00
L3 LED	Split	0.00	3.00
L3 LED	Misclassification	3.00	1.00
L4 Reading Order	ROKT (Kendall's τ)	1.0000	1.0000
L4 Reading Order	ROA	1.0000	1.0000

Perfect Reading Order

Both models achieve ROKT = 1.000 — predicted reading sequence matches ground truth exactly.

Nemotron Leads Classification

86.6% class accuracy vs. 53.0% for DocLayoutYOLO — stronger label discrimination.

DocLayoutYOLO Wins Geometry

Tighter box alignment at 0.806 mean IoU vs. 0.746 for Nemotron-Parse.

Text / OCR Benchmark: Models, Metrics & Results

EasyOCR — Lightweight CRNN-based text parser

Tesseract — Classic LSTM-based OCR engine

CER

S + D + I / N

Character-level edit rate

WER

S + D + I / N

Word-level edit rate

Exact Match

% zones perfect

Zero-edit character match

Model	Corpus CER	Corpus WER	Macro CER	Macro WER	Exact Match
EasyOCR	10.78%	23.08%	**8.06%**	19.73%	31.25%
Tesseract	13.28%	18.32%	11.11%	17.13%	**68.75%**

Interpretation: Tesseract's stronger character segmentation drives a dominant Exact Match Rate (68.75% vs. 31.25%). EasyOCR, however, holds a lower Macro CER (8.06%) — its errors are smaller in magnitude even where matches aren't exact.

Table Extraction Benchmark: Models, Metrics &

Docling — From IBM + Tableformer

TATR — from Microsoft

TEDS (Tree Edit Distance for Structure): normalized similarity between predicted and ground-truth table trees. TEDS-Structural ignores cell text and scores grid topology alone; Grid Exact Match requires every cell to align perfectly.

Model	TEDS Score	TEDS Structural	Grid Exact Match
Docling	**98.16%**	**100.00%**	87.20%
TATR	83.62%	96.58%	25.89%

Interpretation: **Docling** achieves near-perfect structural recovery (100% TEDS-Structural, 98.16% TEDS), showing it robustly reconstructs table grids. TATR's topology looks reasonable in isolation (83.62% TEDS) but collapses under strict cell alignment — only 25.89% Grid Exact Match — revealing that structural similarity and true cell accuracy are not the same thing.

Executive Summary & Next Steps

KEY TAKEAWAYS

- Bug fixes (taxonomy mapping, GT tightness) resolved core measurement distortions from Week 2
- Layout: Nemotron-Parse leads classification (86.6%); DocLayoutYOLO leads geometric tightness (0.806 IoU) — both achieve perfect reading order
- OCR: Tesseract dominates Exact Match (68.75%); EasyOCR holds the lower error magnitude (8.06% Macro CER)
- Tables: EasyOCR delivers near-perfect structural recovery; Tesseract's grid alignment collapses under strict matching

NEXT STEPS

- Re-run DocBank evaluation with adjusted annotation granularity to resolve the structural collapse
- Expand the synthetic catalog dataset — more document types beyond MedCore, wider layout variation
- Re-benchmark on higher-throughput GPU hardware to validate timing and scale results
- Explore task-specific ensembling (e.g., DocLayoutYOLO boxes + Nemotron classification) to combine strengths